PCB Power Budget

9 V DC input feeding an L7805 regulator and nominal 5 V power rail

Power source / load	Part / description	Supply / output voltage	Current budget (mA)	Power / calculation	Status / notes
External power source	Pololu 1470 regulated adapter	9 V DC	1000	9 V × 1 A = 9 W maximum	Rated maximum output: 9 W
Input protection	500 mA fuse	9 V input side	500	Fuse rating: 500 mA	Planned input fuse; BOM does not state DC voltage/interrupt rating
5 V regulator	ST L7805ABV-DG / L7805	9 V in → 5 V out	200	(9 V − 5 V) × 0.2 A = 0.8 W heat	200 mA is a design target, not a measured load
Curiosity Nano board	PIC18F57Q43 + onboard debugger circuitry	5 V rail	100	5 V × 0.1 A = 0.5 W at allocation	100 mA is a provisional planning allocation; actual board current not supplied
MCP6004	Quad op amp	5 V rail	1	1 mA planning allowance	Planning allowance, not measured current
Thermistor divider	10 kΩ NTC + 10 kΩ resistor	5 V rail	0.25	5 V ÷ (10 kΩ + 10 kΩ) = 0.25 mA	Calculated nominal divider current
Red LED	Red LED + 1 kΩ series resistor	5 V rail	3	Approx. (5 V − 2 V) ÷ 1 kΩ = 3 mA	Estimate assumes about 2 V LED forward voltage
Blue LED	Blue LED	5 V rail	TBD	Not calculable from provided data	Supplied schematic/BOM does not show a series current-limiting resistor
External peripherals	Loads connected to headers	5 V rail	TBD	Not calculable from provided data	No external header loads are specified
Partial planning total			104.25		Includes Nano allocation, MCP6004 allowance, divider, and red LED; excludes blue LED and external loads
Adapter maximum power				9 V × 1 A = 9 W	Adapter rating is maximum capacity, not PCB consumption
Regulator heat at 100 mA				(9 V − 5 V) × 0.1 A = 0.4 W	Linear regulator heat estimate
Regulator heat at 200 mA				(9 V − 5 V) × 0.2 A = 0.8 W	Linear regulator heat estimate